



Sabyasachi Samantaray
Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 07/06/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	TSBIE	Sri Chaitanya Jr Kalasala	2021	98.80%
Matriculation	BSET	Sri Chaitanya High School	2019	10

PUBLICATIONS

1. **Breast cancer diagnostics using angular view infrared images** (SPIE, Medical Imaging 2024)
S Samantaray, KS Nair, MFO Baffa, TM Deserno
 - Established the diagnostic potential of **Angular Breast IR Thermograms** using Radiomics and Transfer Learning approaches, achieved best results for oblique images outperforming the diagnostics with frontal images

SCHOLASTIC ACHIEVEMENTS

- Ranked **15th** out of 194 students in **Computer Science Department** based on academic performance ('23)
- Secured an **All India Rank 163** in **Joint Entrance Examination, Mains** out of **11 lakh candidates** ('21)
- Secured an **All India Rank 270** in **Joint Entrance Examination, Advanced** among **2.5 lakh candidates** ('21)
- Secured **State Rank 60** in **TS EAMCET** among **2 lakh+ candidates** conducted by **TSCHE**, Hyderabad ('21)
- Secured an **All India Rank 77** in **VITEEE**, conducted by **Vellore Institute of Technology(VIT)** ('21)
- Among the **top 50+ students** qualified for **OCSC**, Orientation cum Selection Camp having placed for **INPHO** ('21)
- Qualified **NSEC** and **NSEP** having placed in the **National Top 1%** out of **17K+ registered candidates** ('21)
- Recipient of KVPY(SX) Fellowship** organized by **IISC, Bangalore** for securing **All India Rank 587** ('20)
- Secured **Rank 6** in **STSE, Level II** conducted by **ASRAC, Dr. A.S. Rao Awards Council** across AP&TS ('17)

PROFESSIONAL EXPERIENCE

Guided Optimal Extrication

(Summer '23)

Prof. Thomas Deserno | Research Intern, Technische Universität Braunschweig

- Developed a framework to aid in the **automated efficient extrication** of entrapped victims of **Motor Vehicular Accidents**, contributing to the advancement of **3D Digital Rescue Sheet** system using BlenderPy and BeamNG.Tech
- Developed algorithms for **surface detection** and **modelled cost function** for voxelized 3D deformed car meshes
- Explored various traditional **cutting algorithms** and modelled the problem as a **watershed segmentation**

Metabolic Health Risk Analysis

(Spring '23)

Prof. K V Venkatesh | Institute Student Undergraduate Research Project

- Assisted in **Coding the interface** to analyze **risk and lifestyle scores** for various metabolic disorders using blood reports, generated plots depicting distributions of **disease risk** for individuals in healthy and sick categories
- Employed **Sequential Linear Regression** on KEM Diabetic dataset to predict **postprandial glucose values**
- Achieved impressive results with **3.29% average percentage error** in predicting area under the glucose-time curve

TTS for Tagalish

(Dec '23)

Samespace | Internship

- Benchmarked **xTTSv2** against other TTS architectures including ViTS for multiple clients and concurrent queries
- Scripted **audio data collection** from youtube urls using **openai-whisper** ASR, for fine tuning xTTS for Tagalish
- Studied the architecture of xTTS to improve performance throughput at **gRPC server** in absence of batched inference

KEY PROJECTS

Empowering Low Resource Languages | Student Researcher, CSALT Lab, IIT Bombay

(Ongoing)

- Explored **SFT** and **LoRA** methods for instruction fine tuning for low resource languages on multilingual **Mistral**
- Proposed **Phase-Wise Selective Fine Tuning** strategy for overall improvement in translation and understanding
- Extensive literature survey on current research for multilingual language models and bridging of performance gaps

Towards BrainProt4.0 | Student Researcher, Proteomics Lab, IIT Bombay

(Ongoing)

- Using feature selection strategies for **Biomarker Panel Selection** for **Neurological disorders** and **Brain tumors**
- Developing an interactive **protein analysis tool** to aid clinicians assess the druggability of verified human proteins
- Enriching the **Brain Disease Drug Finder (BDDF)** platform, contributing to advancement of **BrainProtv4.0**

ACADEMIC PROJECTS

Image Segmentation

(Spring '23)

Prof. Suyash Awate | Course Project, Medical Image Computing

- Studied, Implemented and Evaluated the **Hybrid Ant Colony Optimisation - K Means** Algorithm for precise Medical Image Segmentation, **outperformed traditional K-Means** for both noisy and low contrast images
- Modified** the proposed **normalised multi-objective** optimization by allowing for minor momentary falls
- Tuned hyper parameters** customized according to the specific characteristics of **diverse organ images**

PCA based Image Reconstruction

(Autumn '22)

Prof. Suyash Awate | Course Project, Data Analysis and Interpretation

- Utilized PCA to analyse image datasets and for their **Dimensionality Reduction** into lower dimensional hyperspace
- Maximised the total dispersion with respect to the chosen hyperplane to **generate new, representative images**
- Reconstructed images** using **Eigen Vector Analysis** and utilized Frobenius Norm to measure closeness

Data Prefetching IPCP2.0

(Spring '23)

Prof. Biswabandan Panda | Course Project, Computer Architecture

- Conducted a **comprehensive evaluation** of the IPCP (**Instruction Pointer Classifier Based Prefetcher**) on various traces and identified areas of improvement specific to different classes of **Instruction Pointers**
- Integrated** all the proposed enhancements to develop a more **advanced** and **efficient version** of the prefetcher
- Demonstrated notable **improvements in IPC values**, with **3.31%** on **Graph traces**, **1.05%** on **SPEC traces**

Cinema A to Z

(Autumn '22)

Prof. Kavi Arya | Course Project, Software Systems

- Deployed** a comprehensive web application using **Django**, providing a **centralized platform** for movie information
- Utilized **Selenium web scraping** for movie related data, effectively handling **Javascript rendered web pages**
- Ideated (M2VE)**Movie-to-Vector-Encoder** using LSTM-based multi-genre classification model and NLP embeddings, leveraged Movie-Vector Embeddings with **K-nearest neighbors** algorithm and cosine similarity for related movies

SimpleFTP

(Spring '23)

Prof. Bhaskaran Raman | Course Project, Computer Networks

- Developed a **client-server program** for **file transfer** after **TCP connection** establishment via Tomlinson handshake
- Extended functionality for **simultaneous multiple client support**, using **select system call** appropriately

Board of Conquest

(Spring '22)

Prof. Rushikesh K. Joshi | Course Project, Abstractions and Paradigms of Programming

- Ideated and Developed** an interactive **PvP**(two player) **board game** in **C++** using **FLTK library** for the **GUI**
- Leveraged principles of **Object-Oriented Programming** and implemented **smoothly functioning** timely updates

Railway Planner

(Autumn '22)

Prof. Supratik Chakraborty | Course Project, Data Structures and Algorithms

- Developed a comprehensive **Journey Planner** for the **Railway Management System**, integrating and implementing functionalities of diverse **Data Structures** for seamless management and swift retrieval of journey information

COURSES UNDERTAKEN

Computer Science	Abstractions & Paradigms, Data Structures & Algorithms, Discrete Structures, Computer Networks, Algorithm Design & Analysis, Logic for Computer Science, Operating Systems, Automata Theory, Medical Image Computing, Foundations of Learning Agents, Advanced ML*, Implementation of Programming Languages*, Database Management Systems*
Mathematics	Calculus - I and II, Linear Algebra, Differential Equations, Distributed Optimization

*To be completed by May '24

TECHNICAL SKILLS

Languages and Tools	C++, Python, MATLAB, Java, VHDL, NS3, Wireshark, GDB, L ^A T _E X, GitHub, Bash
Development	HTML, Bootstrap, Django, Sqlite3, PHP, FLTK, React, Selenium, Doxygen, Sphinx
ML & Data Science	Matplotlib, OpenCV, Tensorflow, Scikit, PyTorch, NLTK, ARIMA, GloVe, Spacy

EXTRACURRICULAR ACHIEVEMENTS

- Member of the **Conference organising team** for **Monsoon Advanced Proteomic School, 2023**, IIT Bombay
- Attended the prestigious **BVM Conference 2023** in **Germany** organised by PLRI for Medical Informatics
- Achieved first place in the **CoDecode** coding Hackathon, hosted by **Techfest, 2022**, as a dynamic team of two
- Won the **Intra SSD Debate Tournament '22** out of 12 competing teams at National Service Scheme, IIT Bombay
- Earned the title of **State Champion** in SIP Abacus'15 competition in **TS&AP** and **Runner-Up** in the **Nationals**